Simulation of DME Production and Separation via Reaction and Distillation in Aspen HYSYS

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Background & Description:

This simulation work focuses on the modeling of a Dimethyl Ether (DME) production process using **Aspen HYSYS**, including both the **reaction stage** and **multistep separation by distillation**. The process involves methanol dehydration over a catalyst to produce DME and water. Due to the presence of unreacted methanol and the formation of water, two distillation columns are required to recover and purify DME and recycle methanol.

Process Description:

A conversion reactor was used to simulate the methanol dehydration reaction:

$$2\ CH3OH \rightarrow CH3OCH3+H2O2$$

The reactor was configured with a **fixed conversion rate** and a **specified outlet temperature of 665 K**, enabling convergence. The reaction feed stream consisted of pure methanol at a flow rate of **575.5 kmol/h**, pressure **1250 kPa**, and temperature **628 K**.

The reactor outlet stream composition was approximately:

- **DME**: 43.0 mol%
- **Water**: 42.2 mol%
- **Unreacted Methanol**: 14.9 mol%

First Distillation Column – DME Separation:

The effluent from the reactor was sent to the first distillation column (Column C1), designed to recover DME as the distillate. This column operates under elevated pressure to allow efficient condensation of DME in vapor phase:

Parameter	**Value**
Condenser Pressure	1000 kPa
Reboiler Pressure	1030 kPa
External Reflux Ratio	1.5
Condenser Temperature	317.8 K (~45 °C)
Reboiler Temperature	443.2 K (~170 °C)
Feed stage	12
Actual trays	22

This allowed nearly complete separation of DME, leaving a methanol/water mixture in the bottoms for further treatment.

Second Distillation Column – MeOH/H_2O Separation (C2):

The second column treats the bottoms from C1 to recover methanol at the top and purge water at the bottom. It was simulated first via shortcut methods, then using a rigorous distillation model.

Feed Composition to C2:

- Methanol: 26.02 mol%
- Water: 73.98 mol%
- Flow rate: 328.6 kmol/h
- Feed temperature: 443.1 K
- Feed pressure: 1030 kPa

Rigorous column configuration:

Parameter	Value
Condenser Pressure	110 kPa
Reboiler Pressure	140 kPa
External Reflux Ratio	3.6
Condenser Temperature	342.2 K (~69 °C)
Reboiler Temperature	381.4 K (~109 °C)
Feed stage	13
Actual trays	21

The simulation showed that a pressure difference of at least 40 kPa between the condenser and the reboiler was required to achieve convergence. Lower pressure gradients led to failure due to inadequate vapor flow or condensation.

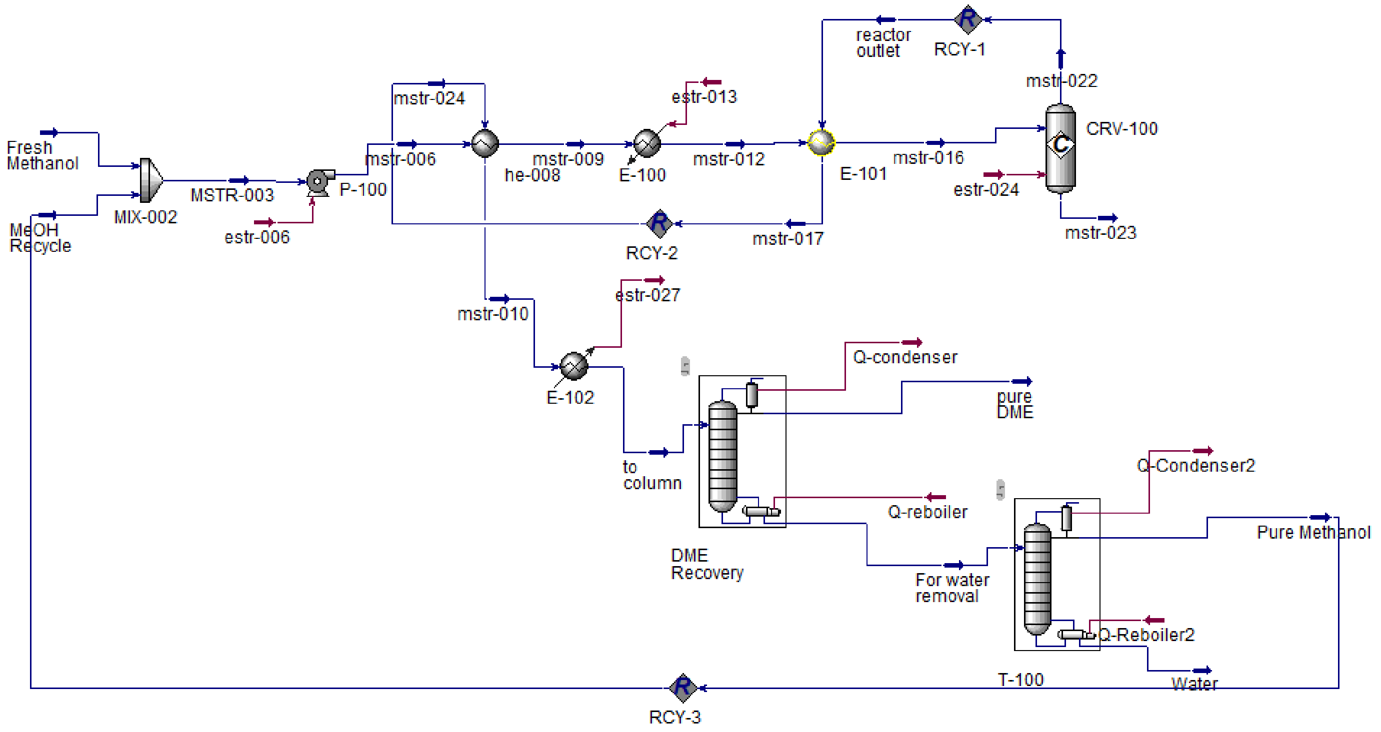
Comparison with Literature:

Compared to the reference process by Luyben (2010), which used a column at 1 atm, 40 trays, and a reflux ratio of 0.5, the present simulation required fewer trays (21) but operated at elevated pressure (110–150 kPa). This change resulted in higher boiling point temperatures, consistent with the thermodynamic behavior of the MeOH–H_2O mixture.

Conclusions:

This simulation demonstrates the feasibility of modeling a complete DME production and separation process using Aspen HYSYS. The combination of shortcut and rigorous distillation modeling allows accurate prediction of column requirements and thermal conditions. The **Peng-Robinson thermodynamic model** provided reliable vapor–liquid equilibrium predictions for the alcohol–ether–water system. Pressure tuning played a critical role in ensuring convergence and operability. The results closely align with theoretical expectations and offer flexibility in design compared to classical configurations.

Flowsheet



Results:

Master Property Tableau

Object	From methanol	MeOH Recycle	Mstr-016	Mstr-022	Mstr-023	Pure DME	Pure methanol	Water
T (K)	320.0	342.2	628.0	665.0	665.0	317.8	342.2	381.4
P (kPa)	101.3	110.0	1250	1250	1250	1000	110.0	140.0
Molar Flow Kgmole/h	500.0	75.47	575.5	575.5	0	248.0	75.47	253.1
Mole Fract (diM-Ether)	0.0100	0.0000	0.0087	0.4297	<0	0.9989	0.0000	0.0000
Mole Fract (H2O)	0.0000	0.0052	0.0007	0.4217	<0	0.0000	0.0052	0.9592
Mole Fract (MEthanol)	0.9900	0.9948	0.9906	0.1486	<0	0.0011	0.9948	0.0408

Nb : Insignificant quantities (<0) of components in the product streams can be considered negligible, effectively approximating them to zero without impacting the overall process results or conclusions (see flowsheet).

References: Luyben, W.L. (2010). *Design and Control of the Methanol to Dimethyl Ether (DME) Process*. Ind. Eng. Chem. Res., 49(13), 6150–6163.